* Java Generates Hashcode ... and the Purpose of
What is the Hashcode in Java and the Purpose of
Generating the Hashcode.

In Java every object gets a hashcode. The hashcode is an integer value generated by the hashcode() method inherited from the object class.

ex:- class Student{

```
Public class Main{
Public static Void main(string[] args){
  Student s1 = new student();
  Student s2 = new student();
System.out.println(s1.hashcode());
System.out.println(s2.hashcode());
} }
```

Output:-

1510467688
1995265320

Ex:- class Student{

```
int id;
Student (int id){
this.id=id;
}
Public int hashcode(){
return id;
} }
```

Purpose of hash code :-

(1) Fast Searching: Collections like Hashmap, Hashset, and Hashtable use hash codes to store and retrieve objects quickly

(2) object Identification:- By default, different objects usually have different hashcodes, helping Java distinguish them

(3) Improves performance: Instead of comparing every object one by one

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82

S	T	U	V	W	X	Y	Z
83	84	85	86	87	88	89	90

Java first compares has codes, making lookups much faster.

The main purpose of generating a hashcode is to enable efficient storage, searching, and retrieval of objects in hash-based collection, such as Hashmap & Hashset.

Eg:- ⇒ The default hashcode() implementation in object is typically based on the objects identity in memory

* if you override equals(), You should usually override hashcode() as well.